

“Electroencephalography **EEG**:



Assist. Proof.

Dr Shereen I. Hajee

Hawler Medical University

College of Medicine

*Department: pharmacology, medical
physics, and clinical biochemistry –*

Med.Physics Unit

Email: shereen.ismail@hmu.edu.krd

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Electroencephalogram (EEG)

Definition

EEG is a **non-invasive neurophysiological technique** that records the **electrical activity of the brain** by placing electrodes on the scalp, regarded as a **powerful tool for brain research and clinical diagnosis**.

What EEG measures

- EEG records **summated postsynaptic potentials** (mainly excitatory and inhibitory)
- Originates primarily from **cortical pyramidal neurons**
- Signals reflect **synchronous neuronal activity**, not single neuron firing



Nature of EEG signals

- Amplitude is very small: **microvolts (μV)**
 - Typical range: **10–100 μV**
- Frequency ranges from **<1 Hz to ~100 Hz**

Why EEG signals are small

- Skull and scalp act as **poor conductors**
- Electrical activity spreads and attenuates before reaching the surface
- Hence signals are recorded in **microvolts**

Advantages of EEG

- **Non-invasive** and safe
- **Excellent temporal resolution** (milliseconds)
- Relatively **low cost**
- Useful for **real-time brain activity monitoring**

Clinical and research uses

- Diagnosis of **epilepsy**
- Sleep studies (**sleep stages**)
- Assessment of **coma and brain death**
- Detection of **encephalopathies**
- Brain–computer interfaces (BCI)
- Cognitive and neuroscience research

How EEG Works

Brain Activity

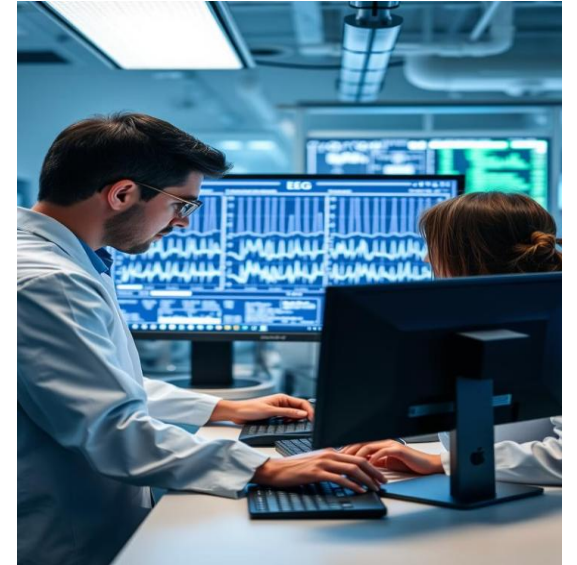
Neurons in the brain communicate with each other using electrical signals.

Electrode Placement

Electrodes are placed on the scalp to pick up these electrical signals.

Signal Amplification

The signals are amplified and recorded by a machine, creating an EEG.



EEG Electrode Placement



10-20 System

The 10-20 system is the standard method for electrode placement.



Electrode Types

EEG uses various electrode types, depending on the specific needs of the study or clinical assessment.



Electrode Locations

Electrodes are placed at specific points on the scalp to capture brain activity in different regions.

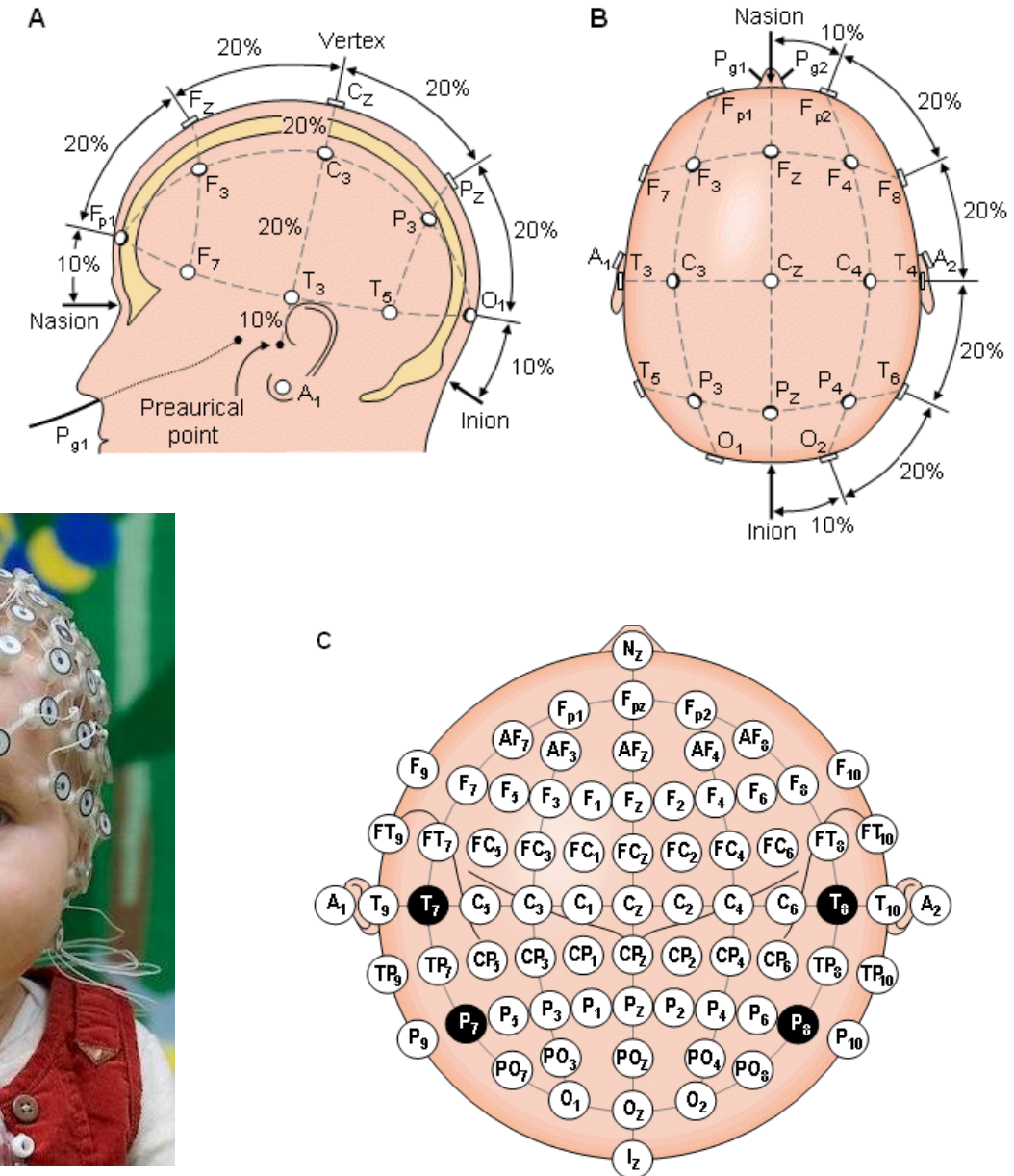


EEG: surface recordings

- International 10/20 or 10/10 system for placing electrodes:

A: earlobes, **C:** central,
P: parietal, **F:** frontal,
O: occipital

- Low impedance 5-10k Ω
- Record **montages**:
 - Bipolar** (electrodes connected to each other)
 - Referential** (electrodes connected to one reference the ears)



Applications of EEG in Research and Clinical Practice

1

Brain-Computer Interfaces

EEG is used to develop brain-computer interfaces (BCIs), which allow people to control devices with their thoughts.

2

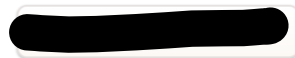
Cognitive Neuroscience

EEG is used in cognitive neuroscience research to study brain activity during cognitive tasks.

3

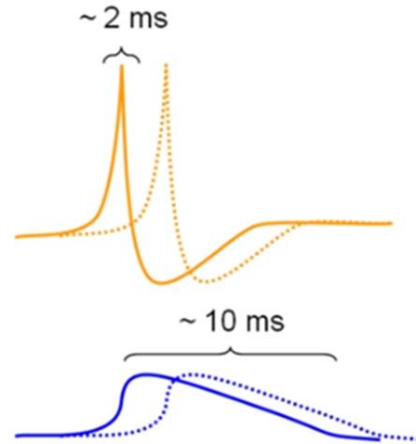
Clinical Diagnosis and Treatment

EEG is used in clinical settings for for diagnosis and treatment of of various brain disorders.



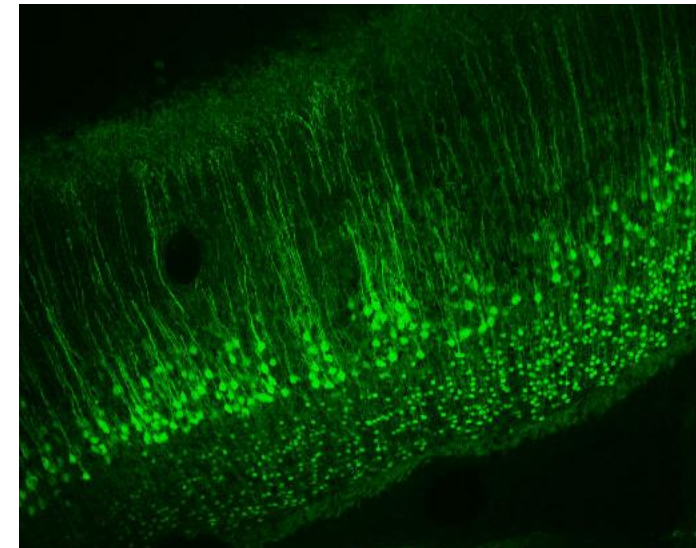
EEG: basis of the signal

- Postsynaptic potentials (PSPs) can be excitatory or inhibitory
- EEG reflects the **summation of synchronous PSPs** across a population of cells, at a point in time.
- **Large pyramidal neurons** in cortex layer V are:
 - ✓ arranged in parallel
 - ✓ similarly-oriented
 - ✓ perpendicular to surface
 - ✓ receive synchronous inputs



Action potentials are biphasic – do not summate

Postsynaptic potentials (PSPs) are monophasic – ideal for summation

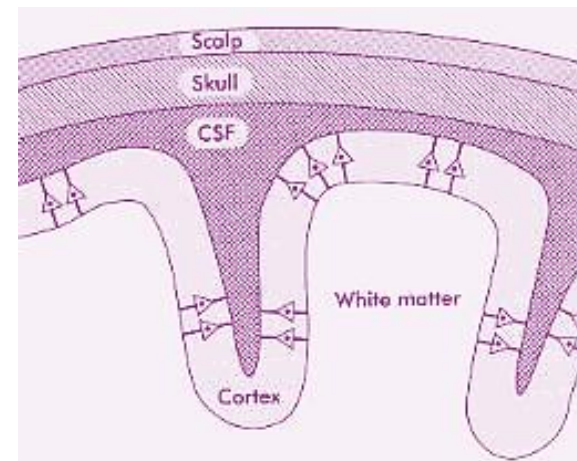
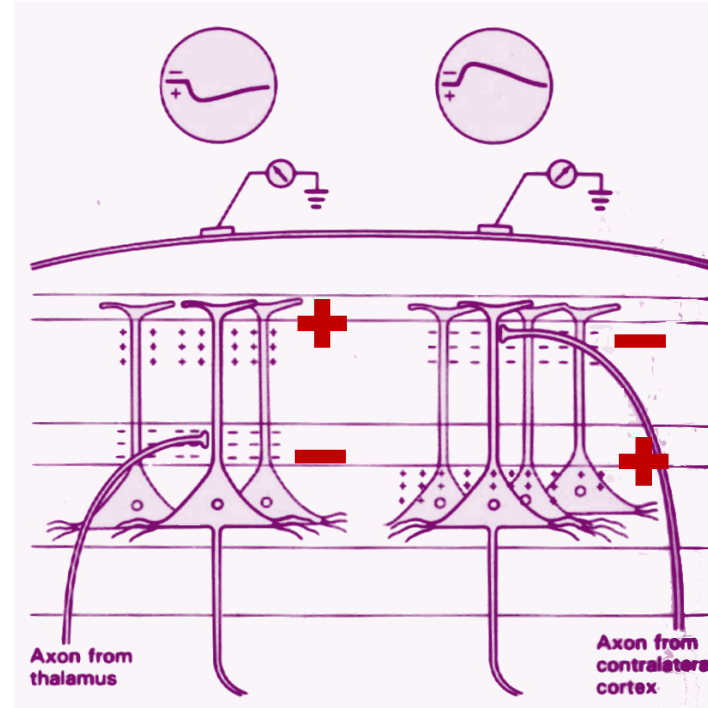


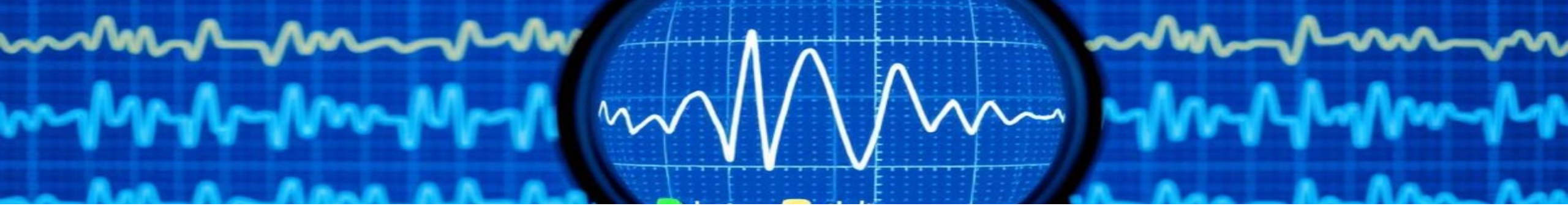
Neurophysiological Basis of EEG

- Single neuron activity is too small to be picked up by EEG
 - EEG reflects the summation of the synchronous activity of many neurons with similar spatial orientations
- Cortical pyramidal neurons produce most of the EEG signal
- Deep sources (subcortical areas) are much more difficult to detect than currents near the skull

EEG: basis of the signal

- Dipole exists between soma and apical dendrites
- Potential behaves as if a **current flow**
- EEG electrodes on scalp detects net **positive** or net **negative** current flow from cortical neurons.





EEG Wave Patterns and Brain Activity



Brain State

Different wave patterns reflect different brain states, such as wakefulness, sleep, or cognitive processes.



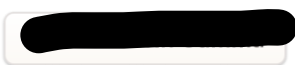
Frequency

EEG waves are categorized by their frequency, measured in Hertz, which indicates the speed of the electrical activity.



Amplitude

The amplitude of the wave indicates the strength of the electrical activity.



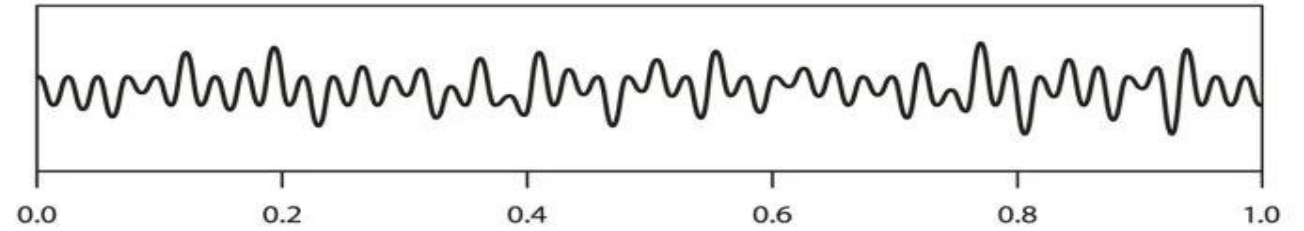
■ EEG Wave Types

Wave	Frequency (Hz)	Amplitude	Physiological State	Clinical / Exam Significance
Delta (δ)	0.5–4 Hz	Very high (up to 100 μ V)	Deep sleep (NREM stage 3–4), unconsciousness	Seen in coma, tumors, brain lesions if present in awake adults
Theta (θ)	4–7 Hz	High	Drowsiness, early sleep (Stage 1–2); normal in children	Abnormal if persistent in awake adults
Alpha (α)	8–13 Hz	Moderate (20–60 μ V)	Awake, relaxed, eyes closed	Blocked by eye opening or mental activity (alpha block)
Beta (β)	13–30 Hz	Low (<20 μ V)	Alertness, active thinking, anxiety	Enhanced by sedatives (benzodiazepines); frontal dominance
Gamma (γ)	30–100 Hz	Very low	High-level cognition, attention, memory, sensory processing	Altered in schizophrenia, Alzheimer's disease, epilepsy research

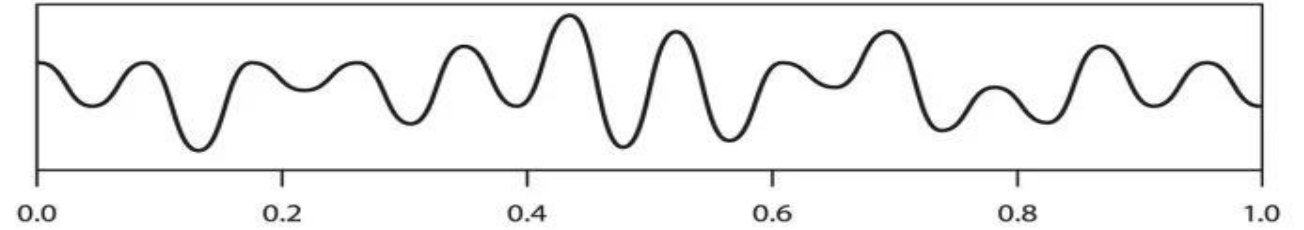
HUMAN BRAIN WAVES

- EEG: frequency spectrum

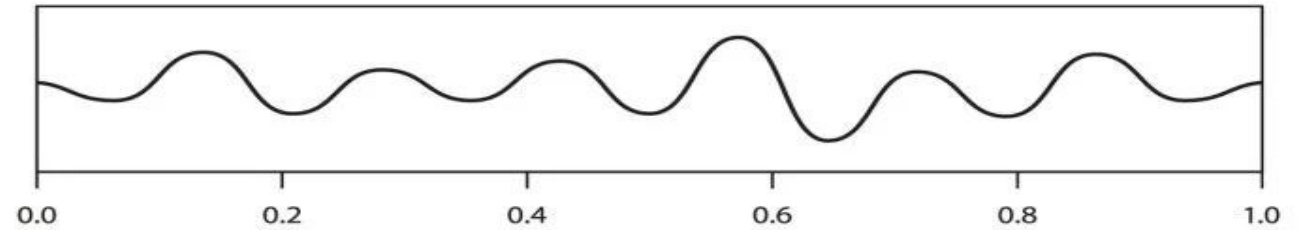
BETA



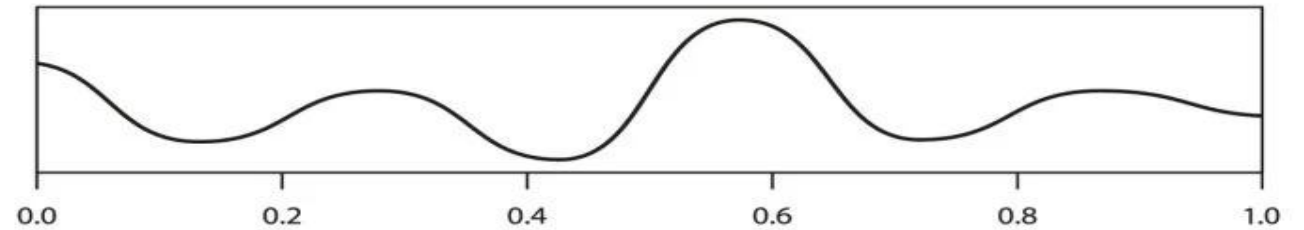
ALPHA



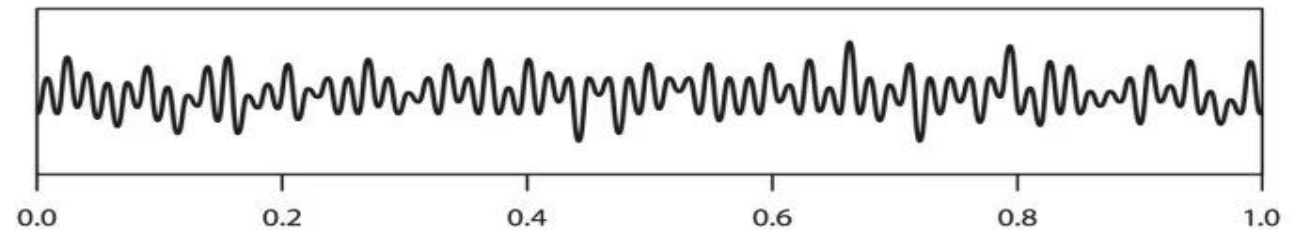
THETA



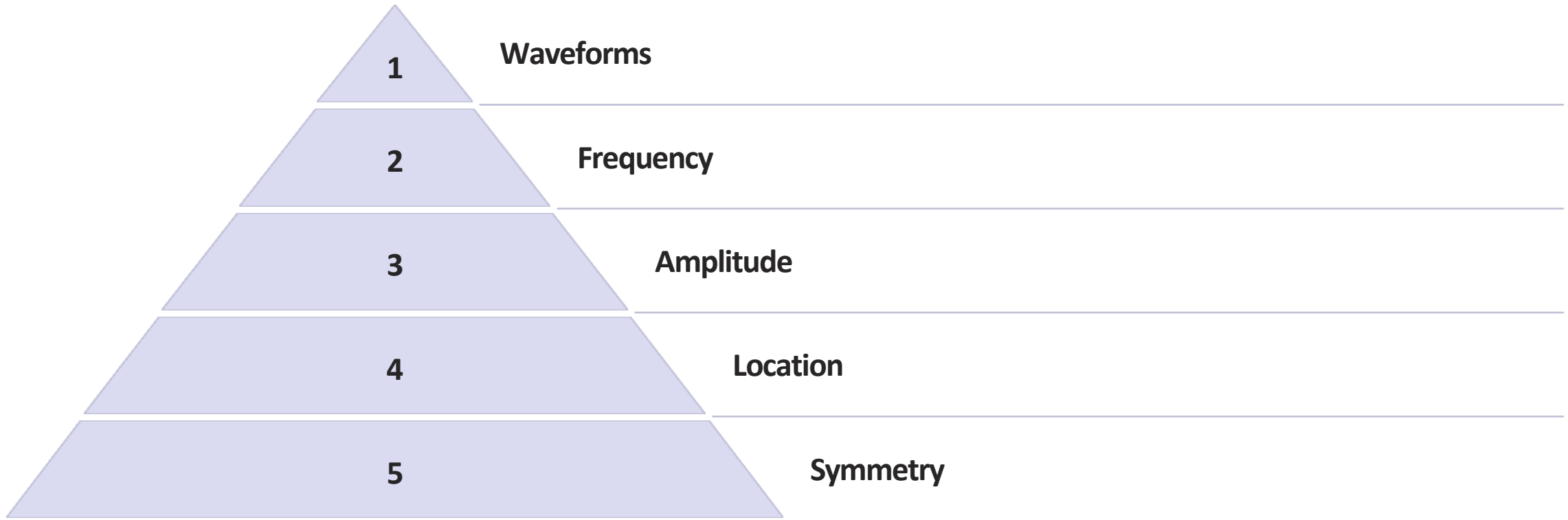
DELTA



GAMMA



Interpreting Normal and Abnormal EEG Waveforms



Interpreting Specific EEG Findings in Clinical Context



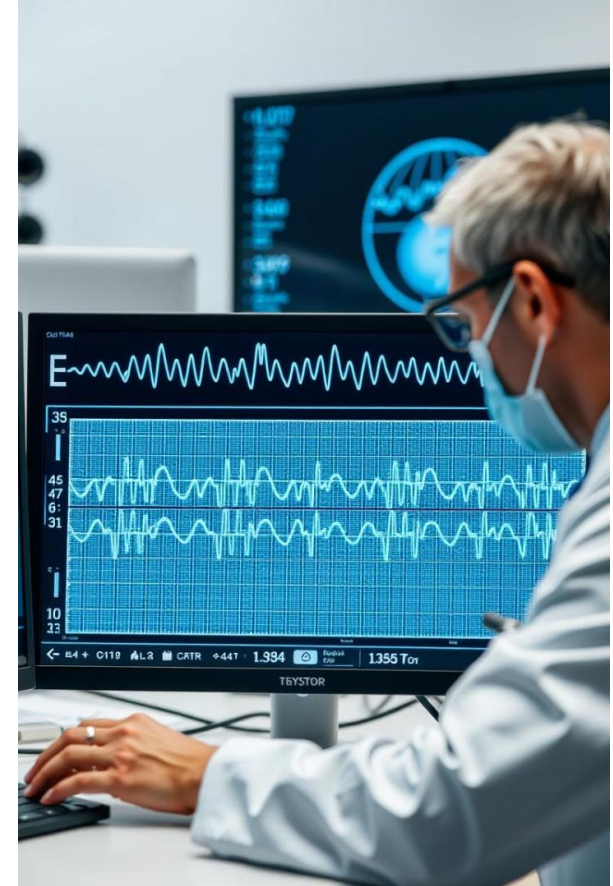
Spikes

Sharp, transient waves suggesting abnormal neuronal activity, often associated with seizures.



Slow Waves

Slow-frequency waves indicating abnormal brain function, often seen in sleep disorders or brain injuries.



Conditions Diagnosed Using EEG

Epilepsy

EEG is the gold standard for diagnosing epilepsy, a disorder that cause seizures.

Sleep Disorders

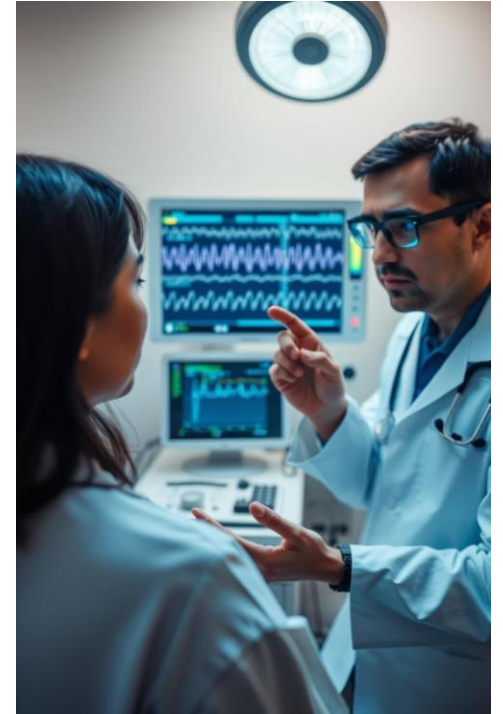
EEG can help diagnose sleep disorders, such as insomnia, narcolepsy, and sleep apnea.

Brain Tumors

EEG can help detect brain tumors by identifying abnormality in brain activity.

Brain Injury

EEG can be used to assess, the severity of brain injury and monitor recovery.



Advantages of EEG

1

Non-Invasive

EEG is a non-invasive procedure, meaning that it does not involve surgery or any other invasive procedures.

2

Safe

EEG is a safe procedure for people of all ages.

3

Affordable

EEG is a relatively affordable diagnostic tool.

4

Widely Available

EEG is widely available in hospitals and clinics.



Recognizing Common EEG Artifacts and Pitfalls

Muscle Activity

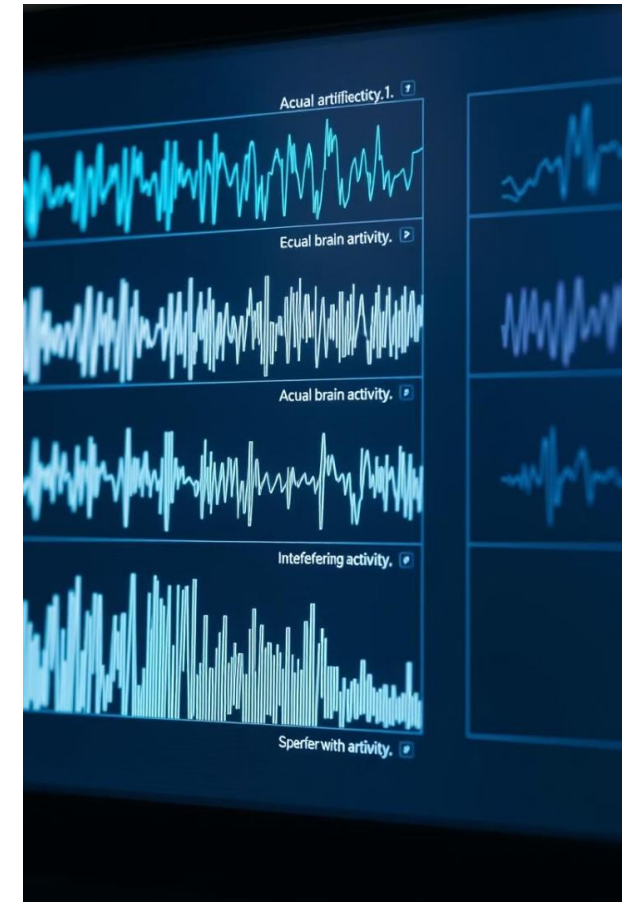
Electrodes picking up muscle contractions, resembling high-frequency activity.

Eye Movements

Electrodes detecting eye movements, producing patterns that can mimic brain waves.

Electrode Displacement

Loose or misplaced electrodes leading to distorted signals or lack of data.



References

1. **Medical Physics, Volume 1: Physical aspects of the Human Body, Harmut Zabel, 2023, 2nd Edition.**
2. **Medical Physics for medical students , Dr Taki A M Al Musawi, (2024),1st Edition.**
3. **Medical physics (John R. Cameron) 1978,1999,2003 2008and 2010**
<https://www.youtube.com/watch?v=XMizSSOejg0>
<https://www.youtube.com/watch?v=1ovv6lmPHSI>
https://www.youtube.com/watch?v=tZcKT4l_JZk
<https://www.youtube.com/watch?v=XMizSSOejg0>